FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Main)-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

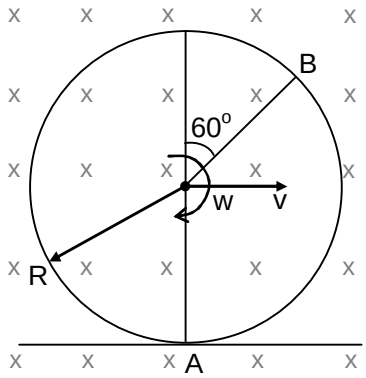
SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. In order to convert a miliammeter of range 1.0 mA and resistance 1.0 ohm into a voltmeter of range 10 V, a resistance of how many ohms should be connected with it and in what manner?
   (A) 999 Ω in series
   (B) 999 Ω in parallel
   (C) 9,999 Ω in series
   (D) 9,999 Ω in parallel

2. The electron emitted in β-radiation originates from :
   (A) inner orbits of atoms
   (B) free electrons existing in nuclei
   (C) decay of a neutron in a nucleus
   (D) photon escaping from the nucleus

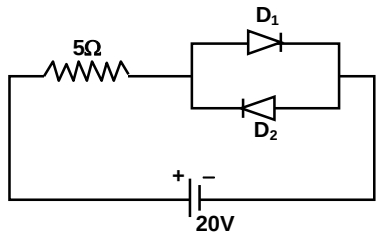
3. A conducting wheel is rolling on the ground in a uniform magnetic field B_o then the emf induced between points A and B; $V_A - V_B$ will be
   (A) $B_o w^2(\sqrt{3}R)$
   (B) $-\frac{\sqrt{3}}{2}Bw^2R$
   (C) $B_o w^2\sqrt{2}R$
   (D) $-B_o w^2\left(\frac{\sqrt{5}}{2}\right)R$



4. Forces acting on a particle moving in a straight line varies with the velocity of the particle as $F = \frac{\alpha}{v}$ where α is constant. The work done by this force in time interval Δt is
   (A) $\alpha\Delta t$
   (B) $\frac{1}{2}\alpha\Delta t$
   (C) $2\alpha\Delta t$
   (D) $\alpha^2\Delta t$

Space for Rough work

5. In the figure shown below, chose the correct answer.
(A) Voltage across D_2 is 20 Volt.
(B) Voltage across D_1 is 20 Volt.
(C) Current through diode D_1 is 4A.
(D) Current through diode D_1 is 0 A.



6. A ferromagnetic substance when heated beyond curie temperature becomes
(A) diamagnetic. (B) ferromagnetic.
(C) nonmagnetic. (D) paramagnetic.

7. The resultant of $\vec{A}$ and $\vec{B}$ is perpendicular to $\vec{A}$. What is the angle between $\vec{A}$ and $\vec{B}$?
(A) $\cos^{-1}\left(\frac{A}{B}\right)$ (B) $\cos^{-1}\left(-\frac{A}{B}\right)$
(C) $\sin^{-1}\left(\frac{A}{B}\right)$ (D) $\sin^{-1}\left(-\frac{A}{B}\right)$

8. In a radioactive reaction an unstable nucleus A dis-integrates into a stable nucleus B. But A is generated at a constant rate of q nucleus per second. Then at steady state number of nucleus of A will be

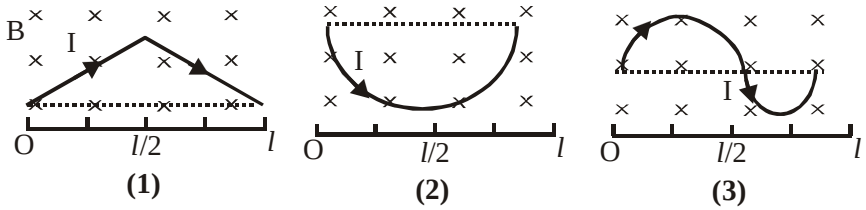
$q\ (\sec^{-1}) \longrightarrow$ A (Parent Nucleus) $\xrightarrow{\lambda}$ B (Daughter Nucleus)

(A) $q\lambda$ (B) $\frac{q}{\lambda}$
(C) $q - \lambda$ (D) $\frac{\lambda}{q}$

9. A block slides down an inclined plane of inclination θ with constant velocity. It is then projected up the plane with an initial speed u. How far up the incline will it move before coming to rest
(A) $\frac{u^2}{4g\sin\theta}$ (B) $\frac{u^2}{g}\sin\theta$
(C) $\frac{u^2}{2g}\sin\theta$ (D) $\frac{u^2}{4g}$

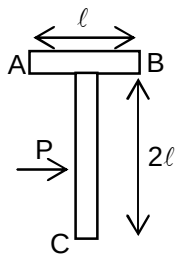
Space for Rough work

10. Three conductors 1, 2, and 3 each carrying the same current I are placed in a uniform magnetic field B, as shown in figure. The forces experienced by conductors 1, 2 and 3 are F_1, F_2 and F_3, respectively



(A) $F_3 > F_2 > F_1$
(B) $F_1 \neq 0$; $F_1 \neq 0$; $F_3 = 0$
(C) F_1 acts upwards, F_2 acts downwards ; $F_3 = 0$
(D) All experience the same force in same direction

11. A T-shaped object with dimensions shown in the figure, is lying on a smooth floor. A force $\vec{F}$ is applied at the point P parallel to AB, such that the object has only the translational motion without rotation. Find the location of P with respect to C.



(A) $\frac{4\ell}{3}$ (B) ℓ
(C) $\frac{2\ell}{3}$ (D) $\frac{3\ell}{2}$

12. A communication earth satellite which takes 24hours to complete one circular orbit eventually has to be replaced by another satellite which has twice the mass of the first. If the new satellite is also to have an orbital period of 24 hr, what is the ratio of radius of the new orbit to the radius of the original orbit?
(A) 1 : 1 (B) 2 : 1
(C) $\sqrt{2}$: 1 (D) 1 : 2

Space for Rough work

13. Imagine a Young's double slit interference experiment performed with wave associated with fast moving electrons produced from an electron gun. The distance between successive maxima will decrease maximum if
(A) the accelerating potential in the electron gun is decreased.
(B) the accelerating potential is increased and the distance of screen from slit is decreased.
(C) the distance of the screen from the slit is increased.
(D) the distance between the slits is decreased.

14. If a person walking at the rate of 3km/hour, the rain appears him to fall vertically. When he increase his speed to 6 km/hr it appears to meet him at angle of 30^0 with vertical. The speed of rain is

(A) $3\sqrt{2}$ km/hr (B) $\frac{3}{\sqrt{2}}$ km/hr

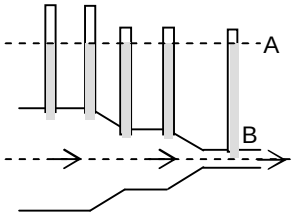
(C) $6\sqrt{2}$ km/hr (D) 6 km/hr

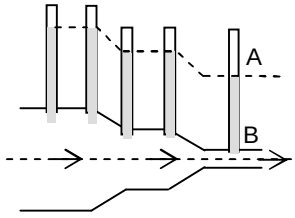
15. A pendulum bob of mass 10^{-2} kg is raised to a height 5×10^{-2}m and then released. At the bottom of its swing, it picks up a mass 10^{-3}kg. To what height will the combined mass rise?
(A) 0.6 m (B) 0.54 m
(C) 0.41 m (D) 0.8 m

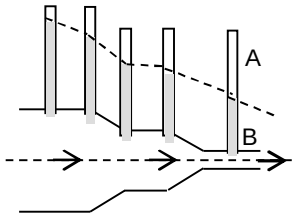
16. A particle performs harmonic motions along y-axis about the origin. The angular frequency is, ω = 4rad/s. At a certain instant the particle has a coordinate y′ = 25 cm and its velocity v′ = 100cm/s. The coordinate y and the velocity of the particle at t = 2.4s after that moment, will be
(A) -28.68 cm, -82.18 cm/s (B) -28.68 cm, 0
(C) 0, -82.18 cm/s (D) 28.68 cm, 82.18 cm/s

Space for Rough work

17. A non-viscous liquid flows through a horizontal pipe of varying cross sectional area. Identify the option which correctly represents the variation of height of rise of liquid in each vertical tube

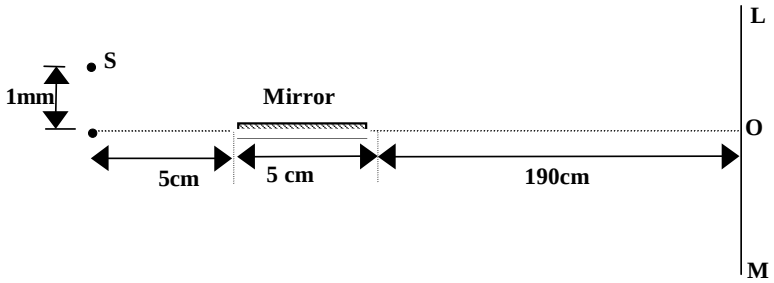
(A) 

(B) 

(C) 

(D) None Of these

18. The arrangement of the Lloyd's mirror experiment is shown in the figure. 'S' is a point source of frequency 6×10^{14} Hz. A and B represent the two ends of a mirror placed horizontally, and LOM represents the screen.



If the number of fringes on the screen is 8x, then x =

(A) 4 (B) 5

(C) 6 (D) 7

Space for Rough work

19. Two narrow cylindrical pipes A and B have the same length. Pipe A is open at both ends and is filled with a monatomic gas of molar mass M_A. Pipe B is open at one end and closed at the other end, and is filled with a diatomic gas of molar mass M_B. Both gases are at the same temperature. If the frequency of the second harmonic in pipe A is equal to the frequency of the third harmonic in pipe B, what is the value of M_A/M_B?

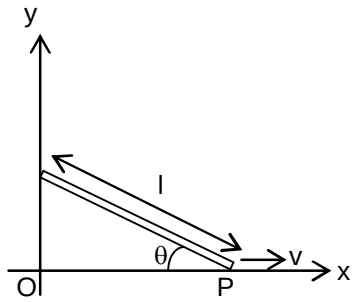
(A) $\frac{100}{189}$ (B) $\frac{200}{189}$

(C) $\frac{400}{189}$ (D) $\frac{500}{189}$

20. The end of a rigid rod of length l is moved with a constant velocity v. Find the speed of the midpoint of the rod

(A) $\frac{v}{2\sin\theta}$ (B) $\frac{v}{2}\tan\theta$

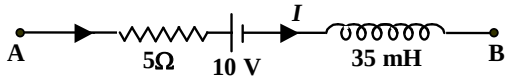
(C) $\frac{v}{2}\cot\theta$ (D) $\frac{v}{2\cos\theta}$



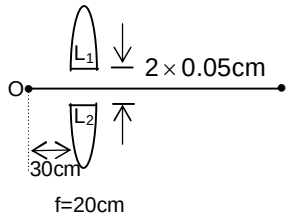
SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. The network shown in Figure is part of a complete circuit. If at a certain instant the current (I) is 5A, and is decreasing at a rate of 10^3 A/s, then $V_B - V_A$ =



22. A point object O is placed at a distance of 0.3 m from a convex lens (focal length 0.2 m) cut into two halves each of which is displaced by 0.0005 m as shown in figure. Find the distance between the images (in mm)



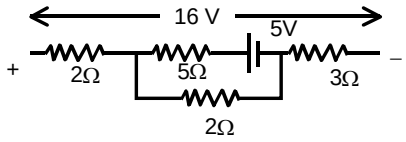
Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. A beam of light wavelength 600 nm from a distant source falls on a single slit 1.00 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance (in mm) between the first dark fringes on either side of the central bright fringe is

24. In the circuit shown below, determine the current through 5V cell



25. A 200 W and a 100 W bulb, both meant for operation at 220 V are connected to a 220 V supply in series. Find total power consumed by them.

Space for Rough work

Chemistry

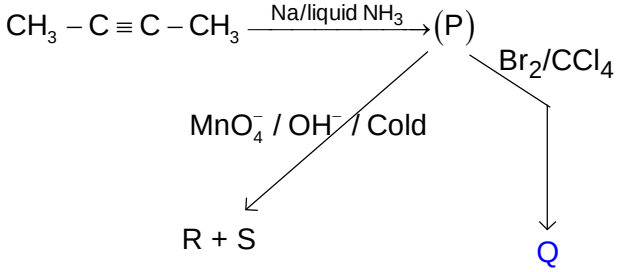
PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

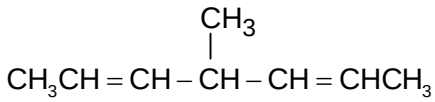
26. The bond order of which molecule is greater than one?
(A) NF_3 (B) BF_3
(C) CF_4 (D) CHF_3

27. Which of the following two compounds of potassium evolve gases when react with water?
(A) KNO_3 and K_2CO_3 (B) KO_2 and K_2S
(C) $KHCO_3$ and KOH (D) K_2O_2 and K_2SO_4

28. 

Which of the following statement is correct about the products of above reaction?
(A) 'Q' contains no asymmetric carbon atom
(B) 'P' and 'Q' are functional isomers
(C) 50% of R and 50% of Q makes a racemic mixture
(D) 'P' has one stereoisomer

Space for Rough work

29. 

How many stereoisomer(s) is/are possible for the above compound?

(A) 2 (B) 3

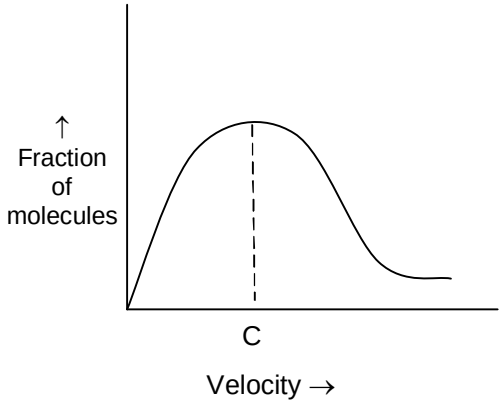
(C) 4 (D) 6

30. What is the wavelength of the motion of the electron of hydrogen atom along the fourth orbit?

[r_0 = Radius of first orbit of 'H' atom]

(A) $2\pi r_0$ (B) $4\pi r_0$

(C) $8\pi r_0$ (D) $16\pi r_0$

31. 

Which of the following velocity is mentioned at C?

(A) $\sqrt{\frac{2RT}{M}}$ (B) $\sqrt{\frac{3RT}{M}}$

(C) $\sqrt{\frac{8RT}{\pi M}}$ (D) $\sqrt{\frac{4RT}{\pi M}}$

Space for Rough work

32. $CO_2(g) + C(s) \rightleftharpoons 2CO(g)$

Which of the following factor cannot affect the product yield of above reaction?
(A) Pressure
(B) Temperature
(C) Catalyst
(D) Addition of inert gas at constant pressure

33. $SrCO_3(s) \rightleftharpoons Sr^{2+}(aq) + CO_3^{2-}(aq)$

Addition of which of the following substance can increase the solubility of $SrCO_3$ according to above reaction?
(A) NaCl
(B) HCl
(C) K_2CO_3
(D) NaOH

34. Neither expansion nor compression work is done in
(A) isothermal process
(B) adiabatic process
(C) isobaric process
(D) isochoric process

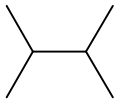
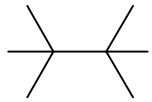
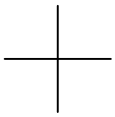
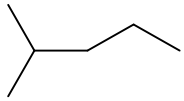
35. Which of the following is not identical between inorganic benzene and organic benzene?
(A) No. of atoms
(B) No. of electrons
(C) No. of π - bonds
(D) No. of polar covalent bonds

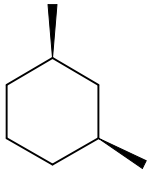
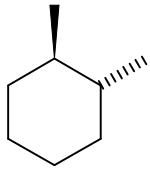
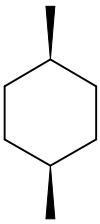
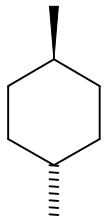
36. Which of the following is the strongest acid?
(A) $CH_3\underset{\underset{Cl}{|}}{C}HCH_2\underset{\underset{F}{|}}{C}HCOOH$
(B) $CH_3\underset{\underset{F}{|}}{C}HCH_2\underset{\underset{Cl}{|}}{C}HCOOH$
(C) $CH_3CH_2\underset{\underset{F}{|}}{C}H\underset{\underset{Cl}{|}}{C}HCOOH$
(D) $CH_3CH_2\underset{\underset{Cl}{|}}{C}H\underset{\underset{F}{|}}{C}HCOOH$

Space for Rough work

37. Which of the following substance on heating produces Glauber's salt?
(A) $NaHCO_3$ (B) $NaHSO_4$
(C) $Na_2S_2O_3$ (D) NaHS

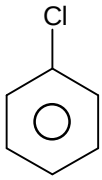
38. Which of the following are heating products of borax?
(A) Na_3B and $NaBO_2$ (B) Na_2O and B_2H_6
(C) $NaBO_2$ and B_2O_3 (D) Na_3B and $B_2O_{3=}$

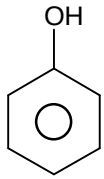
39. Which of the following substance forms the maximum number of monochloro products?
(A)  (B) 
(C)  (D) 

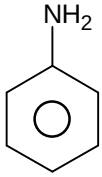
40. Which of the following compounds has maximum number of stereoisomers?
(A)  (B) 
(C)  (D) 

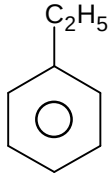
Space for Rough work

41. Which is most reactive towards $CH_2 = CH_2/H^+$?

(A) 

(B) 

(C) 

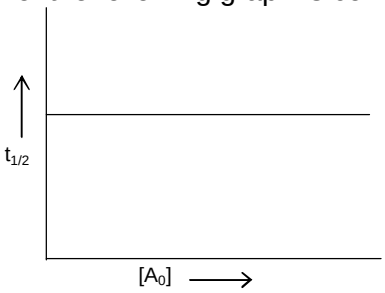
(D) 

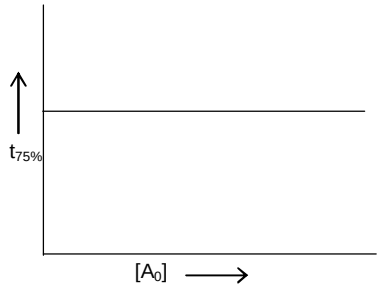
42. Which of the following molecule has square planer shape?

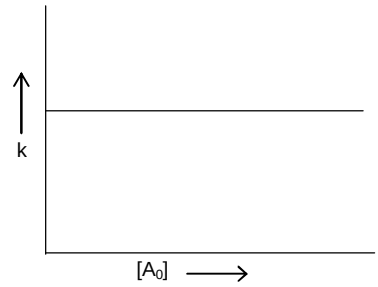
(A) CF_4 (B) SF_4

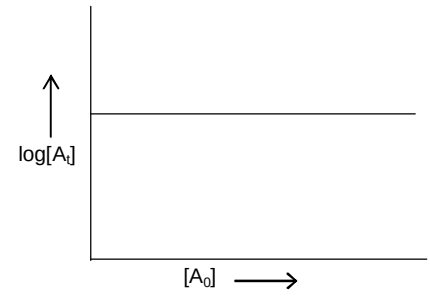
(C) XeF_4 (D) SeF_4

43. Which of the following graph is correct for a zero order reaction?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

44. What is the pH of 10^{-2} M NaH_2PO_4?
[$p^{K_{a_1}}$, $p^{K_{a_2}}$ and $p^{K_{a_3}}$ of H_3PO_4 are 7.2, 10.6 and 12.8]
(A) 9.7 (B) 8.01
(C) 10.6 (D) 8.9

45. In which of the following compound, nitrogen exhibits its lowest oxidation state?
(A) NH_2OH (B) NH_2NH_2
(C) NH_4Cl (D) KNO_3

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. How many of the following species are paramagnetic.
C_2, N_2, O_2, O_2^+, O_2^{2-} F_2, H_2^+, NO

47. How many lone pairs of electrons are present in ICl_2^-.

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. pH of 10^{-3} M solution of acetic acid ($K_a = 1.8 \times 10^{-5}$) is

49. Calculate the spin only magnetic moment of M^{2+}(aq) ion (Z = 27) in BM?

50. If molar mass of the product formed by reaction of phenol with Br_2/Cs_2 is M_o then $m_o/4$ is

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. Let $a > 1$ be a real number and $f(x) = \log_a x^2$ for $x > 0$. If f^{-1} is the inverse function of f and b and c are real numbers then $f^{-1}(b + c)$ is equal to

(A) $f^{-1}(b).f^{-1}(c)$ (B) $f^{-1}(b) + f^{-1}(c)$

(C) $\dfrac{1}{f(b+c)}$ (D) $\dfrac{1}{f^{-1}(b) + f^{-1}(c)}$

52. The function f (x) = $\dfrac{\log(\pi + x)}{\log(e + x)}$ $(x \geq 0)$ is

(A) increasing in [0, ∞)

(B) decreasing in [0, ∞)

(C) increasing in $\left[0, \dfrac{\pi}{e}\right)$ and decreasing in $\left[\dfrac{\pi}{e}, \infty\right)$

(D) decreasing in $\left[0, \dfrac{\pi}{e}\right)$ and increasing in $\left[\dfrac{\pi}{e}, \infty\right)$

53. Locus of all point P (x, y) satisfying $x^3 + y^3 + 3xy = 1$ consists of union of

(A) a line and an isolated point (B) a line pair and an isolated point

(C) a line and a circle (D) a circle and an isolated point

Space for Rough work

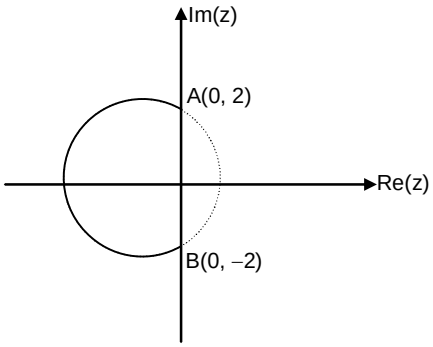
54. If $\tan\theta_1$, $\tan\theta_2$, $\tan\theta_3$ and $\tan\theta_4$ are the roots of the equation $x^4 - x^3\sin 2\beta + x^2\cos 2\beta - x\cos\beta - \sin\beta = 0$ then $\tan\left(\theta_1 + \theta_2 + \theta_3 + \theta_4\right)$ is equal to

(A) $\sin\beta$ (B) $\cos\beta$

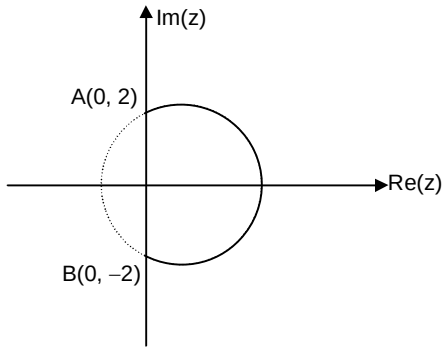
(C) $\tan\beta$ (D) $\cot\beta$

55. If $\arg\left(\frac{z-2i}{z+2i}\right) = \frac{\pi}{6}$, then which of the following options gives the correct locus of z ?

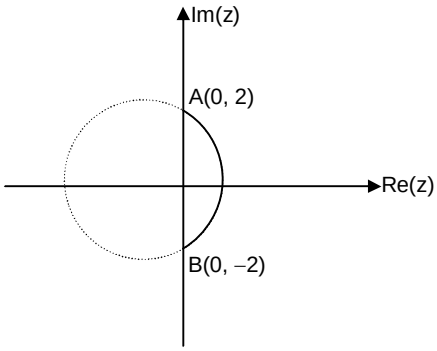
(A)



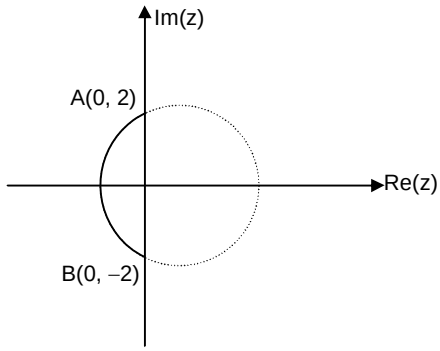
(B)



(C)



(D)



Space for Rough work

56. If $I = \int \sin x.\cos x.\cos 2x.\cos 4x.\cos 8x.\cos 16x\,dx$, then the value of I is

(A) $\frac{\sin 16x}{1024} + c$ (B) $-\frac{\cos 32x}{1024} + c$

(C) $\frac{\cos 32x}{1096} + c$ (D) $-\frac{\cos 32x}{1096} + c$

57. $y = f(x)$ is a function which satisfies:

(i) $f(0) = 0$

(ii) $f''(x) = f'(x)$ and

(iii) $f'(0) = 1$

then the area bounded by the graph of $y = f(x)$, the lines $x = 0, x - 1 = 0$ and $y + 1 = 0$, is:

(A) e (B) $e - 2$

(C) $e - 1$ (D) $e + 1$

58. Let z_1, z_2, z_3 be three distinct points $|z| = 1$. If θ_1, θ_2 and θ_3 be the arguments of z_1, z_2, z_3 respectively then $\cos(\theta_1 - \theta_2) + \cos(\theta_2 - \theta_3) + \cos(\theta_3 - \theta_1)$

(A) $\geq -\frac{3}{2}$ (B) $\leq -\frac{3}{2}$

(C) $\geq \frac{3}{2}$ (D) ≤ 2

59. If the function $f(x) = -4e^{\frac{1-x}{2}} + 1 + x + \frac{x^2}{2} + \frac{x^3}{3}$ and $g(x) = f^{-1}(x)$, then the value of $g'\left(-\frac{7}{6}\right)$ equals

(A) $\frac{1}{5}$ (B) $-\frac{1}{5}$

(C) $\frac{6}{7}$ (D) $-\frac{6}{7}$

Space for Rough work

60. If α,β are two real numbers satisfying $\alpha^2+\beta^2=5$ and $3\left(\alpha^5+\beta^5\right)=11\left(\alpha^3+\beta^3\right)$, then $\alpha\beta$ is:

(A) 2 (B) 1
(C) 7 (D) 9

61. Two teachers are interviewing top 6 students in **FTRE 2016** exam in two different subjects, starting at the same time. Each teacher interviews a student for 15 minutes. The number of ways in which the interviews can be scheduled is

(A) 6! (B) 5!
(C) 44 x 6! (D) 265 x 6!

62. The length of the perpendicular from the origin to the plane passing through the point $\vec{a}$ and containing the line $\vec{r}=\vec{b}+\lambda\vec{c}$ is

(A) $\dfrac{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}{\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|}$ (B) $\dfrac{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}{\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}\right|}$

(C) $\dfrac{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}{\left|\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|}$ (D) $\dfrac{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}{\left|\vec{c}\times\vec{a}+\vec{a}\times\vec{b}\right|}$

63. If $A=\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$, then $\lim\limits_{n\to\infty}\dfrac{A^n}{n}$ is (where $\theta\in R$)

(A) a zero matrix (B) an identity matrix
(C) $\begin{bmatrix}0 & 1\\ 0 & -1\end{bmatrix}$ (D) $\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}$

Space for Rough work

64. A line is drawn from the point P (1, 1, 1) and perpendicular to a line with direction ratios 1, 1, 1 to intersect the plane $x + 2y + 3z = 4$ at Q. The locus of Q is

(A) $\frac{x}{1} = \frac{y-5}{-2} = \frac{z+2}{1}$ (B) $\frac{x}{-2} = \frac{y-5}{1} = \frac{z+2}{1}$

(C) $x = y = z$ (D) $\frac{x}{2} = \frac{y}{3} = \frac{2}{5}$

65. The lines $2x^2 - 3y^2 + xy - 3x - 7y + 2 = 0$ represent two normals of a circle which passes through the focus of the parabola $(5x + 15)^2 + \left(5y - \frac{5}{2}\right)^2 = \left(3x + 4y - \frac{1}{2}\right)^2$. The equation of the circle is

(A) $4x^2 + 4y^2 - 8x - 8y - 65 = 0$ (B) $4x^2 + 4y^2 - 8x + 8y - 65 = 0$

(C) $x^2 + y^2 - 2x - 2y - 73 = 0$ (D) $x^2 + y^2 - 2x + y + 73 = 0$

66. The asymptotes of the hyperbola $y = x + \frac{1}{x}$ are

(A) $y = 0, y = x$ (B) $y = 0, y = -x$

(C) $x = 0, y = -x$ (D) $x = 0, y = x$

67. If $x\frac{dy}{dx} = y(\log_e y - \log_e x + 1)$, then the solution of the equation is

(A) $y \log_e\left(\frac{x}{y}\right) = cx$ (B) $x \log_e\left(\frac{y}{x}\right) = cy$

(C) $\log_e\left(\frac{y}{x}\right) = c\,x$ (D) $\log_e\left(\frac{x}{y}\right) = cy$

Space for Rough work

68. $\frac{1}{4}+\frac{1(3)}{4(6)}+\frac{1(3)(5)}{4(6)(8)}+.....\infty$

(A) 1 (B) 2
(C) 3 (D) 4

69. The minimum value of $(\tan A-\cos B)^2+(\cot A-\sin B)^2$ for all permissible independent variables A and B is

(A) 0 (B) $\sqrt{2}-1$
(C) $3-2\sqrt{2}$ (D) cannot be determined

70. The constant term in the expansion of $\left(x^2+\frac{1}{x^2}+y+\frac{1}{y}\right)$ is:

(A) 4900 (B) 4950
(C) 5050 (D) 5151

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. The total number of solutions of $|\cot x|=\cot x+\frac{1}{\sin x}$, $x\in[0, 3\pi]$ is

72. Cosine of the acute angle between the curves $y=3^{x-1}\ln x$ and $y=x^x-1$ is

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. Let $T = \int_0^{\ln 2} \frac{2e^{3x} + e^{2x} - 1}{e^{3x} + e^{2x} - e^{x} + 1} dx$, then e^T equals:

74. Three parallel chords of a circle have lengths 2,3,4 and subtend angles $\alpha, \beta, \alpha + \beta$ at the centre respectively (given $\alpha + \beta < \pi$) **,** then $1000\cos\alpha$ is equal to

75. From a set of numbers $\{1,2,3,.....,999\}$, six distinct numbers are chosen. Out of these six numbers three are chosen and labeled as a_1, a_2, a_3. The remaining are labeled as b_1, b_2, b_3. The probability that a brick of dimensions $a_1 \times a_2 \times a_3$ fits in a box of dimensions $b_1 \times b_2 \times b_3$ is equal to (assume that the brick can be rotated before putting into the box)

Space for Rough work